

Test 1:

Duration for Fused: 53:40

llmsys_hw4 >  mt_fused_err										
306	Tra:	~/11868/llmsys_hw4	• Contains emphasized items	157	[51:08<02:46,	20.78s/it,	loss=5.11,	lr=0.002,	tokens_per_sec=235]	
307	Training (epoch 0 / 1):	95%		149/157	[51:28<02:46,	20.78s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=245]	
308	Training (epoch 0 / 1):	96%		150/157	[51:28<02:24,	20.68s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=245]	
309	Training (epoch 0 / 1):	96%		150/157	[51:49<02:24,	20.68s/it,	loss=4.94,	lr=0.002,	tokens_per_sec=245]	
310	Training (epoch 0 / 1):	96%		151/157	[51:49<02:03,	20.59s/it,	loss=4.94,	lr=0.002,	tokens_per_sec=245]	
311	Training (epoch 0 / 1):	96%		151/157	[52:08<02:03,	20.59s/it,	loss=5.17,	lr=0.002,	tokens_per_sec=251]	
312	Training (epoch 0 / 1):	97%		152/157	[52:08<01:41,	20.38s/it,	loss=5.17,	lr=0.002,	tokens_per_sec=251]	
313	Training (epoch 0 / 1):	97%		152/157	[52:29<01:41,	20.38s/it,	loss=4.86,	lr=0.002,	tokens_per_sec=243]	
314	Training (epoch 0 / 1):	97%		153/157	[52:29<01:21,	20.45s/it,	loss=4.86,	lr=0.002,	tokens_per_sec=243]	
315	Training (epoch 0 / 1):	97%		153/157	[52:50<01:21,	20.45s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=237]	
316	Training (epoch 0 / 1):	98%		154/157	[52:50<01:01,	20.63s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=237]	
317	Training (epoch 0 / 1):	98%		154/157	[53:11<01:01,	20.63s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=240]	
318	Training (epoch 0 / 1):	99%		155/157	[53:11<00:41,	20.70s/it,	loss=5.04,	lr=0.002,	tokens_per_sec=240]	
319	Training (epoch 0 / 1):	99%		155/157	[53:32<00:41,	20.70s/it,	loss=4.97,	lr=0.002,	tokens_per_sec=232]	
320	Training (epoch 0 / 1):	99%		156/157	[53:32<00:20,	20.95s/it,	loss=4.97,	lr=0.002,	tokens_per_sec=232]	
321	Training (epoch 0 / 1):	99%		156/157	[53:40<00:20,	20.95s/it,	loss=5.22,	lr=0.002,	tokens_per_sec=172]	
322	Training (epoch 0 / 1):	100%		157/157	[53:40<00:00,	16.85s/it,	loss=5.22,	lr=0.002,	tokens_per_sec=172]	
323	Training (epoch 0 / 1):	100%		157/157	[53:40<00:00,	20.51s/it,	loss=5.22,	lr=0.002,	tokens_per_sec=172]	

Duration for non-Fused: 1:01:26

llmsys_hw4 >  mt_non-fused_37970684.err										
303	Training (epoch 0 / 1):	94%		147/157	[58:21<03:39,	21.92s/it,	loss=4.62,	lr=0.002,	tokens_per_sec=220]	
304	Training (epoch 0 / 1):	94%		148/157	[58:21<03:19,	22.14s/it,	loss=4.62,	lr=0.002,	tokens_per_sec=220]	
305	Training (epoch 0 / 1):	94%		148/157	[58:43<03:19,	22.14s/it,	loss=4.84,	lr=0.002,	tokens_per_sec=233]	
306	Training (epoch 0 / 1):	95%		149/157	[58:43<02:55,	21.94s/it,	loss=4.84,	lr=0.002,	tokens_per_sec=233]	
307	Training (epoch 0 / 1):	95%		149/157	[59:05<02:55,	21.94s/it,	loss=4.82,	lr=0.002,	tokens_per_sec=220]	
308	Training (epoch 0 / 1):	96%		150/157	[59:05<02:35,	22.17s/it,	loss=4.82,	lr=0.002,	tokens_per_sec=220]	
309	Training (epoch 0 / 1):	96%		150/157	[59:27<02:35,	22.17s/it,	loss=4.68,	lr=0.002,	tokens_per_sec=237]	
310	Training (epoch 0 / 1):	96%		151/157	[59:27<02:11,	21.84s/it,	loss=4.68,	lr=0.002,	tokens_per_sec=237]	
311	Training (epoch 0 / 1):	96%		151/157	[59:50<02:11,	21.84s/it,	loss=4.92,	lr=0.002,	tokens_per_sec=215]	
312	Training (epoch 0 / 1):	97%		152/157	[59:50<01:51,	22.27s/it,	loss=4.92,	lr=0.002,	tokens_per_sec=215]	
313	Training (epoch 0 / 1):	97%		152/157	[1:00:11<01:51,	22.27s/it,	loss=4.62,	lr=0.002,	tokens_per_sec=237]	
314	Training (epoch 0 / 1):	97%		153/157	[1:00:11<01:27,	21.92s/it,	loss=4.62,	lr=0.002,	tokens_per_sec=237]	
315	Training (epoch 0 / 1):	97%		153/157	[1:00:34<01:27,	21.92s/it,	loss=4.82,	lr=0.002,	tokens_per_sec=219]	
316	Training (epoch 0 / 1):	98%		154/157	[1:00:34<01:06,	22.21s/it,	loss=4.82,	lr=0.002,	tokens_per_sec=219]	
317	Training (epoch 0 / 1):	98%		154/157	[1:00:55<01:06,	22.21s/it,	loss=4.85,	lr=0.002,	tokens_per_sec=234]	
318	Training (epoch 0 / 1):	99%		155/157	[1:00:55<00:43,	21.94s/it,	loss=4.85,	lr=0.002,	tokens_per_sec=234]	
319	Training (epoch 0 / 1):	99%		155/157	[1:01:17<00:43,	21.94s/it,	loss=4.75,	lr=0.002,	tokens_per_sec=224]	
320	Training (epoch 0 / 1):	99%		156/157	[1:01:17<00:22,	22.05s/it,	loss=4.75,	lr=0.002,	tokens_per_sec=224]	
321	Training (epoch 0 / 1):	99%		156/157	[1:01:26<00:22,	22.05s/it,	loss=4.96,	lr=0.002,	tokens_per_sec=153]	
322	Training (epoch 0 / 1):	100%		157/157	[1:01:26<00:00,	17.89s/it,	loss=4.96,	lr=0.002,	tokens_per_sec=153]	
323	Training (epoch 0 / 1):	100%		157/157	[1:01:26<00:00,	23.48s/it,	loss=4.96,	lr=0.002,	tokens_per_sec=153]	

Speedup : 61/53 = 1.15

Test 1:

Duration for Fused: 32:42

llmsys_hw4 > mt_fused.err									
301	Training (epoch 0 / 1):	95%		140/157	[30:45<02:47, 13.27s/it,	loss=4.90, lr=0.002, tokens_per_sec=393]			
302	Training (epoch 0 / 1):	94%		147/157	[30:45<02:24, 14.50s/it,	loss=4.96, lr=0.002, tokens_per_sec=393]			
303	Training (epoch 0 / 1):	94%		147/157	[30:57<02:24, 14.50s/it,	loss=4.86, lr=0.002, tokens_per_sec=413]			
304	Training (epoch 0 / 1):	94%		148/157	[30:57<02:04, 13.78s/it,	loss=4.86, lr=0.002, tokens_per_sec=413]			
305	Training (epoch 0 / 1):	94%		148/157	[31:10<02:04, 13.78s/it,	loss=5.11, lr=0.002, tokens_per_sec=407]			
306	Training (epoch 0 / 1):	95%		149/157	[31:10<01:46, 13.33s/it,	loss=5.11, lr=0.002, tokens_per_sec=407]			
307	Training (epoch 0 / 1):	95%		149/157	[31:22<01:46, 13.33s/it,	loss=5.04, lr=0.002, tokens_per_sec=412]			
308	Training (epoch 0 / 1):	96%		150/157	[31:22<01:30, 12.97s/it,	loss=5.04, lr=0.002, tokens_per_sec=412]			
309	Training (epoch 0 / 1):	96%		150/157	[31:34<01:30, 12.97s/it,	loss=4.94, lr=0.002, tokens_per_sec=419]			
310	Training (epoch 0 / 1):	96%		151/157	[31:34<01:15, 12.66s/it,	loss=4.94, lr=0.002, tokens_per_sec=419]			
311	Training (epoch 0 / 1):	96%		153/157	[31:47<01:15, 12.66s/it,	loss=5.17, lr=0.002, tokens_per_sec=388]			
312	Training (epoch 0 / 1):	97%		152/157	[31:47<01:03, 12.72s/it,	loss=5.17, lr=0.002, tokens_per_sec=388]			
313	Training (epoch 0 / 1):	97%		152/157	[31:59<01:03, 12.72s/it,	loss=4.86, lr=0.002, tokens_per_sec=411]			
314	Training (epoch 0 / 1):	97%		153/157	[31:59<00:50, 12.55s/it,	loss=4.86, lr=0.002, tokens_per_sec=411]			
315	Training (epoch 0 / 1):	97%		153/157	[32:12<00:50, 12.55s/it,	loss=5.04, lr=0.002, tokens_per_sec=387]			
316	Training (epoch 0 / 1):	98%		154/157	[32:12<00:37, 12.66s/it,	loss=5.04, lr=0.002, tokens_per_sec=387]			
317	Training (epoch 0 / 1):	98%		154/157	[32:24<00:37, 12.66s/it,	loss=5.04, lr=0.002, tokens_per_sec=411]			
318	Training (epoch 0 / 1):	99%		155/157	[32:24<00:25, 12.51s/it,	loss=5.04, lr=0.002, tokens_per_sec=411]			
319	Training (epoch 0 / 1):	99%		155/157	[32:37<00:25, 12.51s/it,	loss=4.97, lr=0.002, tokens_per_sec=382]			
320	Training (epoch 0 / 1):	99%		156/157	[32:37<00:12, 12.67s/it,	loss=4.97, lr=0.002, tokens_per_sec=382]			
321	Training (epoch 0 / 1):	99%		156/157	[32:42<00:12, 12.67s/it,	loss=5.22, lr=0.002, tokens_per_sec=265]			
322	Training (epoch 0 / 1):	100%		157/157	[32:42<00:00, 10.29s/it,	loss=5.22, lr=0.002, tokens_per_sec=265]			
323	Training (epoch 0 / 1):	100%		157/157	[32:42<00:00, 12.50s/it,	loss=5.22, lr=0.002, tokens_per_sec=265]			

Duration for non-Fused: 45:41

llmsys_hw4 > mt_non-fused_37973860.err									
297	Training (epoch 0 / 1):	92%		144/157	[42:19<03:53, 17.94s/it,	loss=4.78, lr=0.002, tokens_per_sec=311]			
298	Training (epoch 0 / 1):	92%		145/157	[42:19<03:28, 17.38s/it,	loss=4.78, lr=0.002, tokens_per_sec=311]			
299	Training (epoch 0 / 1):	92%		145/157	[42:37<03:28, 17.38s/it,	loss=4.57, lr=0.002, tokens_per_sec=281]			
300	Training (epoch 0 / 1):	93%		146/157	[42:37<03:12, 17.50s/it,	loss=4.57, lr=0.002, tokens_per_sec=281]			
301	Training (epoch 0 / 1):	93%		146/157	[42:56<03:12, 17.50s/it,	loss=4.75, lr=0.002, tokens_per_sec=262]			
302	Training (epoch 0 / 1):	94%		147/157	[42:56<02:59, 17.97s/it,	loss=4.75, lr=0.002, tokens_per_sec=262]			
303	Training (epoch 0 / 1):	94%		147/157	[43:13<02:59, 17.97s/it,	loss=4.62, lr=0.002, tokens_per_sec=296]			
304	Training (epoch 0 / 1):	94%		148/157	[43:13<02:38, 17.64s/it,	loss=4.62, lr=0.002, tokens_per_sec=296]			
305	Training (epoch 0 / 1):	94%		148/157	[43:30<02:38, 17.64s/it,	loss=4.84, lr=0.002, tokens_per_sec=294]			
306	Training (epoch 0 / 1):	95%		149/157	[43:30<02:19, 17.44s/it,	loss=4.84, lr=0.002, tokens_per_sec=294]			
307	Training (epoch 0 / 1):	95%		149/157	[43:48<02:19, 17.44s/it,	loss=4.82, lr=0.002, tokens_per_sec=273]			
308	Training (epoch 0 / 1):	96%		150/157	[43:48<02:03, 17.69s/it,	loss=4.82, lr=0.002, tokens_per_sec=273]			
309	Training (epoch 0 / 1):	96%		150/157	[44:05<02:03, 17.69s/it,	loss=4.68, lr=0.002, tokens_per_sec=290]			
310	Training (epoch 0 / 1):	96%		151/157	[44:05<01:45, 17.56s/it,	loss=4.68, lr=0.002, tokens_per_sec=290]			
311	Training (epoch 0 / 1):	96%		151/157	[44:23<01:45, 17.56s/it,	loss=4.92, lr=0.002, tokens_per_sec=286]			
312	Training (epoch 0 / 1):	97%		152/157	[44:23<01:27, 17.53s/it,	loss=4.92, lr=0.002, tokens_per_sec=286]			
313	Training (epoch 0 / 1):	97%		152/157	[44:40<01:27, 17.53s/it,	loss=4.62, lr=0.002, tokens_per_sec=285]			
314	Training (epoch 0 / 1):	97%		153/157	[44:40<01:10, 17.53s/it,	loss=4.62, lr=0.002, tokens_per_sec=285]			
315	Training (epoch 0 / 1):	97%		153/157	[44:57<01:10, 17.53s/it,	loss=4.82, lr=0.002, tokens_per_sec=302]			
316	Training (epoch 0 / 1):	98%		154/157	[44:57<00:51, 17.24s/it,	loss=4.82, lr=0.002, tokens_per_sec=302]			
317	Training (epoch 0 / 1):	98%		154/157	[45:14<00:51, 17.24s/it,	loss=4.85, lr=0.002, tokens_per_sec=292]			
318	Training (epoch 0 / 1):	99%		155/157	[45:14<00:34, 17.20s/it,	loss=4.85, lr=0.002, tokens_per_sec=292]			
319	Training (epoch 0 / 1):	99%		155/157	[45:34<00:34, 17.20s/it,	loss=4.75, lr=0.002, tokens_per_sec=247]			
320	Training (epoch 0 / 1):	99%		156/157	[45:34<00:18, 18.11s/it,	loss=4.75, lr=0.002, tokens_per_sec=247]			
321	Training (epoch 0 / 1):	99%		156/157	[45:41<00:18, 18.11s/it,	loss=4.96, lr=0.002, tokens_per_sec=184]			
322	Training (epoch 0 / 1):	100%		157/157	[45:41<00:00, 14.71s/it,	loss=4.96, lr=0.002, tokens_per_sec=184]			
323	Training (epoch 0 / 1):	100%		157/157	[45:41<00:00, 17.46s/it,	loss=4.96, lr=0.002, tokens_per_sec=184]			

Speedup : 45/32=1.4